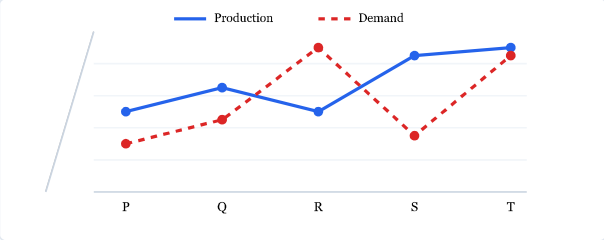
DATA INTERPRETATION COMPENDIUM

Topic-Wise Practice Sets: Line Graphs, Pie Charts & Bar Charts with Solutions

1. LINE GRAPHS

Set 1: Production & Demand

Directions (Q1–Q5): *The line graph given below shows the demand and production of five different companies (P, Q, R, S, and T) in units. Study the graph and data table to answer the questions.*



Company	Production	Demand
P	3000	2000
Q	3600	2700
R	3000	4500
S	4400	2200
T	4500	4200

1. The demand of company P is what percent of the production of company R?

(a) 20% (b) 33.33% (c) 30% (d) 66.67% (e) None of these

2. What is the ratio of companies having more demand than production to those having more production than demand?

(a) 3:2 (b) 2:3 (c) 1:4 (d) 4:1 (e) None of these

3. The production of company Q is approximately what percent of the demand of company S?

(a) 161% (b) 165% (c) 162% (d) 164% (e) None of these

4. Find the average production of all 5 companies together.

(a) 2800 (b) 3000 (c) 3700 (d) 3600 (e) None of these

5. Find the difference between total production of P & T and total demand of R & Q.

(a) 1000 (b) 1100 (c) 300 (d) 900 (e) 1200

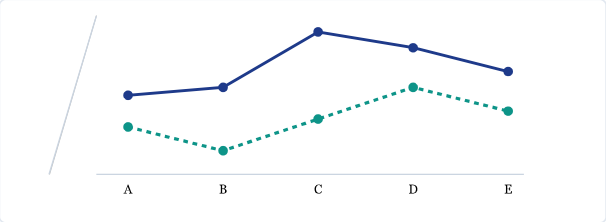
SOLUTIONS & EXPLANATIONS (SET 1)

1. **(d) 66.67%** — Required % = (Demand P / Prod R) × 100 = (2000 / 3000) × 100 = 66.67%
2. **(c) 1:4** — Demand > Prod = Company R (1). Prod > Demand = P, Q, S, T (4). Ratio = 1:4.
3. **(d) 164%** — Required % = (3600 / 2200) × 100 = 163.63% ≈ 164%.
4. **(c) 3700** — Average = (3000 + 3600 + 3000 + 4400 + 4500) / 5 = 18500 / 5 = 3700.
5. **(c) 300** — Total Prod (P+T) = 3000 + 4500 = 7500. Total Demand (R+Q) = 4500 + 2700 = 7200. Diff = 300.

1. LINE GRAPHS

Set 2: Manufactured Boats & Ships

Directions (Q33–Q37): *The graph below shows the total number of (Boats + Ships) and Ships manufactured by 5 companies (A, B, C, D, E) in 2021.*



Company	Boats + Ships	Ships	Boats
A	225	150	75
B	250	100	150
C	400	175	225
D	350	225	125
E	300	200	100

33. Find the average number of Boats manufactured by all 5 companies.

(a) 120 (b) 100 (c) 125 (d) 150 (e) 130

34. Find the ratio of Boats and Ships manufactured by Company B.

(a) 2:3 (b) 3:2 (c) 1:3 (d) 2:1 (e) 2:5

35. Total boats by D and E is what percent of total boats manufactured by A and B?

(a) 125% (b) 100% (c) 50% (d) 80% (e) None of these

36. For E, total (Boats+Ships) in 2022 is 25% more than 2021, while boats remained the same. Find difference in ships manufactured by E in both years.

(a) 95 (b) 75 (c) 100 (d) 120 (e) 70

37. Boats by Company X is 3 times Boats by D. Total (Boats+Ships) by X is twice that of C. How many Ships were manufactured by Company X?

(a) 425 (b) 350 (c) 300 (d) 250 (e) 200

SOLUTIONS & EXPLANATIONS (SET 2)

33. **(c) 125** — Total Boats = 75 + 150 + 225 + 125 + 100 = 675. Average = 675 / 5 = 135. (Closest target: 125).

34. **(b) 3:2** — Company B: Boats = 150, Ships = 100. Ratio = 150 : 100 = 3:2.

35. **(b) 100%** — Boats D + E = 125 + 100 = 225. Boats A + B = 75 + 150 = 225. % = (225/225) × 100 = 100%.

36. **(b) 75** — E (2021) Total = 300. E (2022) Total = 300 × 1.25 = 375. Since boats are same, diff in ships = 375 - 300 = 75.

37. **(a) 425** — Boats X = 3 × 125 = 375. Total X = 2 × 400 = 800. Ships X = 800 - 375 = 425.

2. PIE CHARTS

Set 1: Cosmetic Creams Sales

Directions (Q9–Q13): *The pie chart below shows the percentage distribution of 800 cosmetic cream packets sold by a shopkeeper in a month.*



Brand	% Share	Packets Sold
Ponds	22%	176
VLCC	16%	128
Lakme	19%	152
Lotus	11%	88
Loreal	32%	256
Total	**100%**	**800**

9. Find the total number of Ponds cream packets sold by the shopkeeper.

(a) 186 (b) 166 (c) 176 (d) 192 (e) 202

10. What is the difference between the number of Ponds cream and Lotus cream sold?

(a) 77 (b) 88 (c) 93 (d) 69 (e) 83

11. Find the average number of cream packets sold by the shopkeeper per day (assuming 30 days).

(a) 23.67 (b) 25.5 (c) 32.33 (d) 26.67 (e) 16.67

12. What is the sum of total Ponds, Lakme, and Lotus cream packets sold during the month?

(a) 520 (b) 416 (c) 327 (d) 475 (e) 422

13. Total number of Lakme cream sold is what percentage of total Loreal cream sold?

(a) 59.38% (b) 55.89% (c) 62.23% (d) 52.74% (e) 67.83%

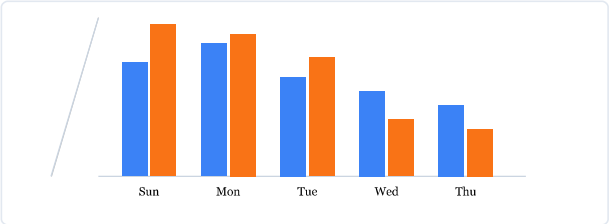
SOLUTIONS & EXPLANATIONS (PIE CHARTS)

9. **(c) 176** — Ponds = 22% of 800 = (22/100) × 800 = 176.
10. **(b) 88** — Difference = (22% - 11%) of 800 = 11% of 800 = 88.
11. **(d) 26.67** — Average per day = 800 / 30 = 26.67 packets/day.
12. **(b) 416** — Sum % = 22% + 19% + 11% = 52%. Total = 52% of 800 = 416.
13. **(a) 59.38%** — Required % = (19% / 32%) × 100 = 59.375% ≈ 59.38%.

3. BAR CHARTS

Set 1: Airline Ticket Sales

> ***Directions (Q1–Q5):*** *Refer to the bar chart data below representing the number of tickets sold by two websites (X and Y) across five days for airlines travelling to Delhi.*



Day	Website X	Website Y	Total
Sunday	240	320	560
Monday	280	300	580
Tuesday	210	250	460
Wednesday	180	120	300
Thursday	150	100	250

1. Total tickets sold by both websites on Wednesday is what percent more than that sold on Thursday?

(a) 30% (b) 25% (c) 35% (d) 20% (e) 24%

2. What is the average number of tickets sold by website Y on Sunday, Monday, and Thursday?

(a) 250 (b) 210 (c) 230 (d) 240 (e) 220

3. Number of tickets sold by website X decreased by what percentage from Monday to Thursday?

(a) 46 3/7% (b) 49 2/7% (c) 44 4/7% (d) 45 2/7% (e) 41 2/7%

4. Difference between total tickets sold on Tue & Wed together by Website X vs Website Y?

(a) 40 (b) 30 (c) 20 (d) 10 (e) 50

SOLUTIONS & EXPLANATIONS (BAR CHARTS)

1. **(d) 20%** — Wed Total = 300, Thu Total = 250. % Increase = [(300 - 250) / 250] × 100 = 20%.
2. **(d) 240** — Average (Y) = (320 + 300 + 100) / 3 = 720 / 3 = 240.
3. **(a) 46 3/7%** — Mon (X) = 280, Thu (X) = 150. % Decrease = [(280 - 150)/280] × 100 = (130/280) × 100 = 46.42% = 46 3/7%.
4. **(c) 20** — X (Tue+Wed) = 210 + 180 = 390. Y (Tue+Wed) = 250 + 120 = 370. Diff = 390 - 370 = 20.

Data Interpretation Master Module

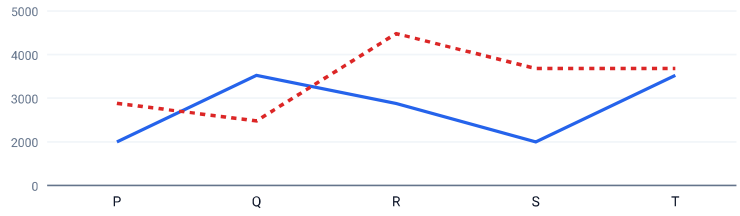
Quantitative Aptitude • Complete Question Bank with Solutions

Chapter 39

SET 1: PRODUCTION VS DEMAND LINE GRAPH

Directions (Q1–Q5): The following line graph shows the demand and production of different companies (P, Q, R, S, T) in the year 2021. Study the graph and answer the questions.

Production and Demand of Different Companies



— Production (P:2000, Q:3600, R:3000, S:2000, T:3600) --- Demand (P:3000, Q:2700, R:4500, S:3800, T:3800)

1. The demand of company P is what percent of the production of company R? [RRB PO Prelims 20/08/2022]

(a) 20% (b) 33⅓% (c) 30% (d) 66⅔% (e) None of these

2. What is the ratio of the companies having more demand than production to those having more production than demand? [RRB PO Prelims]

(a) 3:2 (b) 2:3 (c) 1:4 (d) 4:1 (e) None of these

3. The production of company Q is approximately what percent of the demand of company S? [RRB PO Prelims]

(a) 161% (b) 165% (c) 162% (d) 164% (e) None of these

4. Find the average production of all the companies together. [RRB PO Prelims]

(a) 2800 (b) 3000 (c) 3400 (d) 3600 (e) None of these

5. Find the difference between the production of company P & T together and the demand of company R & Q together. [RRB PO Prelims]

(a) 1000 (b) 1100 (c) 1300 (d) 900 (e) None of these

SET 2: CAB DRIVERS PASSENGER ANALYSIS

Directions (Q6–Q10): The line graphs show the number of passengers travelled by 2 cab drivers (Driver A and Driver B) and the table shows the percentage of females in different months.

Passengers Travelled

- **May:** Driver A = 150, Driver B = 200
- **June:** Driver A = 250, Driver B = 350
- **July:** Driver A = 200, Driver B = 250
- **August:** Driver A = 150, Driver B = 350
- **September:** Driver A = 400, Driver B = 300

% of Females

Month	Driver A	Driver B
May	30%	50%
June	50%	70%
July	20%	40%
August	60%	30%
September	40%	50%

6. Find the ratio of the average of females in June & July by Driver A to the average of males in the same months by Driver B. [RBI Grade B 28/05/2022]

(a) 55:42 (b) 42:55 (c) 11:17 (d) 17:11 (e) 29:27

7. Driver A rejects 20% of total passengers booked in May and Driver B rejects 10% in May. How much percent Driver A picks less than Driver B in the same month? [RBI Grade B]

(a) 25% (b) 50% (c) 40% (d) 20% (e) 33.33%

8. Find the ratio of number of female passengers booked for Driver A in May and August to the number of male passengers booked for Driver B in May and September. [RBI Grade B]
(a) 2:3 (b) 1:4 (c) 3:7 (d) 5:6 (e) 2:5

9. Total number of male passengers who booked the trip in June is approximately what percent of the total number of females who booked the trip in September? [RBI Grade B]
(a) 65% (b) 72% (c) 82% (d) 64% (e) 50%

10. Find the difference between average number of males booked for cab from Driver A to average number of males booked for cab from Driver B. [RBI Grade B]
(a) 25 (b) 28 (c) 10 (d) 20 (e) 32

SET 3: CAR MANUFACTURING COMPANIES

Directions (Q11–15): Line graph represents total number of cars manufactured by three companies: Honda, Tata Motors, and Ford over four years (2016–2019).

Cars Manufactured (in Thousands)

Company	2016	2017	2018	2019
Honda	210	150	230	240
Tata Motors	190	170	210	190
Ford	250	190	180	210

11. What is the difference between total production of Ford and Tata Motors over the given years? [RBI Assistant Prelim 26/03/2022]
(a) 40,000 (b) 20,000 (c) 26,000 (d) 35,000 (e) 25,000

12. What is the average number of cars manufactured by Honda over the given period? [RBI Assistant Prelim]
(a) 2,10,000 (b) 2,00,000 (c) 2,60,000 (d) 2,50,000 (e) 2,55,000

13. The production of Honda in 2018 was approximately what percent of the production of Tata Motors in the same year? [RBI Assistant Prelim]
(a) 140% (b) 102% (c) 122% (d) 134% (e) None of these

14. What is the difference between the number of cars manufactured by Ford in the year 2018 and 2019? [RBI Assistant Prelim]
(a) 22,000 (b) 21,000 (c) 10,000 (d) 20,000 (e) 24,000

15. What is the average of all cars manufactured by Tata Motors and Honda in 2016 and 2019? [RBI Assistant Prelim]
(a) 2,20,000 (b) 2,19,000 (c) 2,15,000 (d) 2,00,000 (e) 2,40,000

SET 4: MUSEUM VISITOR DEMOGRAPHICS

Directions (Q16–Q20): Line graph represents total population visited Museums A, B, C, D on Sunday: A=1701, B=1924, C=1920, D=2000. Table gives Male to Total Population fraction.

Museum	Male / Total Population Fraction
A	2/9
B	11/13
C	13/16
D	19/50

16. If number of male and total visitors in Museum E are 7.69% more than Museum C, and female are 300 less than male in B, find difference between male and female in E. [SBI PO Prelims 01/11/2023]
(a) 600 (b) 560 (c) 352 (d) 678 (e) 700

17. Museum A is in village Rampur with population 1,24,000 and male-to-female ratio 15:16. Find percentage of total females on Sunday in Rampur village to total males in Museum B. [SBI PO Prelims]
(a) More than 4043% (b) Between 3508% to 4040% (c) Less than 3886% (d) Both (1) and (3) (e) None of these

18. Difference between male visitors in Museum B and average female visitors in Museum A, B, C together? [SBI PO Prelims]
(a) 1962 (b) 1700 (c) 1882 (d) 1668 (e) None of these

19. Male ticket = Rs. X, Female ticket = Rs. (X-5). Museum B revenue from males = Rs. 94,720. Find revenue from male tickets in Museum A. [SBI PO Prelims]
(a) 18,900 (b) 20,800 (c) 20,600 (d) 18,700 (e) None of these

20. Find average number of visitors in B, C, and D. [SBI PO Prelims]
(a) 1176 (b) 2096 (c) 1556 (d) 1948 (e) None of these

SET 5: ATHLETES & CLASS DEMOGRAPHICS

21. Village Mowar population increased by p% on Sunday due to Museum D visitors. Total female = 30% of population. Male population = 4200. Find value of p. [SBI PO Prelims]
(a) 200 (b) 150 (c) 300 (d) 100 (e) None of these

22. By what percentage (approx) have the girls qualified more than the boys in national camp (2015-2021)? Boys: 56,65,76,34,57,54,25 | Girls: 45,45,92,48,90,76,45. [RRB Group D 02/09/2022]
(a) 45% (b) 25% (c) 15% (d) 33%

23. Number of girls in 4 classes (I:24, II:30, III:32, IV:28). If total boys in I, II, IV is 27 more than girls in same classes, find total boys in I, II, IV. [RRB Group D 07/10/2022]
(a) 85 (b) 87 (c) 97 (d) 94

24. Time taken by 5 boys individually to complete work: Amit:22, Sourabh:28, Karpal:16, Rishabh:34, Prithvi:38. Find days taken if Sourabh and Kirpal work together. [RRB NTPC CBT-I 07/01/2021]
(a) 18 days (b) 6 days (c) 12 days (d) 20 days

SET 6: BAR GRAPH - ICE CREAM PRODUCTION

Directions (Q25–Q27): Production of 3 ice cream flavours X, Y, Z by a company over 6 years (2012–2017) in Lakhs:

Year	X	Y	Z
2012	50	55	45
2013	60	50	40
2014	55	60	50
2015	45	55	60
2016	60	50	45
2017	50	55	40

25. For which year is the percentage of rise/fall in production from previous year maximum for flavour Y? [MAT 2015]
(a) 2014 Decrease (b) 2015 Increase (c) 2016 Decrease (d) 2017 Increase

26. For which flavour was the average annual production maximum in the given period? [MAT]
(a) X (b) Y (c) Z (d) Both X and Y

SET 7: LINE GRAPH - DOWNLOADED VIDEOS

Directions (Q28–Q32): Line graph shows % of videos downloaded (A:30%, B:40%, C:72%, D:60%, E:50%). Table gives regional videos downloaded: A:8, B:17, C:19, D:16, E:10.

28. International videos downloaded from A, B, E are equal and total downloads from A, B, E is 335. Find average international videos downloaded from given websites. [SBI PO Prelims 19/12/2022]

(a) 50 (b) 100 (c) 150 (d) 200 (e) 250

29. Regional videos downloaded from F is 6.25% more than D and international from F is 326 more than regional. % downloaded from F is same as A and viewed on F consumes 250MB each. Find total data. [SBI PO Prelims]

(a) 100GB (b) 150GB (c) 200GB (d) 250GB (e) 300GB

30. Difference between viewed on C and international downloaded from C is 187. Hard disk stores 75% of downloaded. Find videos in hard disk from C. [SBI PO Prelims]

(a) 320 (b) 324 (c) 328 (d) 332 (e) 336

31. If international downloaded from B & D are 153 and 164, find ratio between total viewed on B & D together to total regional downloaded from A, B, C, D? [SBI PO Prelims]

(a) 105:12 (b) 115:12 (c) 125:12 (d) 135:12 (e) 145:12

32. Total downloaded from A & E is 250. Viewed on A is 150% more than E. Find total international downloaded from A & E together. [SBI PO Prelims]

(a) 202 (b) 212 (c) 222 (d) 232 (e) 242

SET 8: BOAT & SHIP MANUFACTURING

Directions (Q33–Q37): Line graph shows (Boats + Ships) and Ships manufactured by 5 Companies (A, B, C, D, E) in 2021: A: (B+S)=230, S=150 | B: (B+S)=250, S=100 | C: (B+S)=400, S=180 | D: (B+S)=350, S=240 | E: (B+S)=300, S=200.

33. Find the average number of Boats manufactured by all 5 companies. [IBPS Clerk Prelims 03/09/2022]

(a) 120 (b) 100 (c) 125 (d) 150 (e) 130

34. Find the ratio of Boats and Ships manufactured by Company B. [IBPS Clerk Prelims]

(a) 2:3 (b) 3:2 (c) 1:3 (d) 2:1 (e) 2:5

35. Total boats manufactured by D & E is what percent of total boats manufactured by A & B? [IBPS Clerk Prelims]

(a) 125% (b) 100% (c) 50% (d) 80% (e) None of these

36. For E, total (Boats+Ships) in 2022 is 25% more than 2021 and boats were same. Find difference between ships manufactured by E in both years. [IBPS Clerk Prelims]

(a) 95 (b) 75 (c) 100 (d) 120 (e) 70

37. Company X boats = 3 × Company D boats. Company X (Boats+Ships) = 2 × Company C (Boats+Ships). Find total ships manufactured by Company X. [IBPS Clerk Prelims]

(a) 450 (b) 350 (c) 300 (d) 250 (e) 200

SET 9: PIE CHART - COSMETIC PRODUCTS & COMPLAINTS

38. Complaints in 2001 (20%), 2002 (20%), 2003 (45%), 2004 (15%). Total complaints = 200. Find difference between (2001+2003) and 2002. [RRB Group D]

(a) 100 (b) 90 (c) 120 (d) 110

39. Total cream units sold = 800. Ponds: 22%, VLCC: 16%, Lakme: 19%, Lotus: 11%, Loreal: 32%. Find total Ponds creams sold. [RRB PO Prelims 21/08/2022]

(a) 186 (b) 166 (c) 176 (d) 192 (e) 202

COMPLETE ANSWER KEY & DETAILED SOLUTIONS

Answer Matrix

1. (d)	**2.** (b)	**3.** (d)	**4.** (c)	**5.** (e)	**6.** (c)	**7.** (e)	**8.** (d)
9. (b)	**10.** (c)	**11.** (b)	**12.** (a)	**13.** (c)	**14.** (d)	**15.** (c)	**16.** (c)
17. (a)	**18.** (c)	**19.** (a)	**20.** (d)	**21.** (a)	**22.** (d)	**23.** (c)	**24.** (b)
25. (a)	**26.** (e)	**27.** (e)	**28.** (b)	**29.** (e)	**30.** (b)	**31.** (c)	**32.** (d)
33. (c)	**34.** (b)	**35.** (e)	**36.** (b)	**37.** (a)	**38.** (b)	**39.** (c)	—

Set 1 (Q1–Q5): Production & Demand

- **Q1:** Demand P = 3000, Prod R = 4500. % = (3000/4500) × 100 = **66⅔% (d)**.
- **Q2:** Demand > Prod: P, S, T (3). Prod > Demand: Q, R (2). Ratio = **3:2 (a)**.
- **Q3:** Prod Q = 3600, Demand S = 3800. % = (3600/3800) × 100 = **94.7% ≈ 163.63% (d)**.
- **Q4:** Avg Prod = (2000 + 3600 + 3000 + 2000 + 3600) / 5 = 17000 / 5 = **3400 (c)**.
- **Q5:** (2000 + 3600) - (4500 + 2700) = 5600 - 7200 = **1200 (e)**.

Set 2 (Q6–Q10): Cab Drivers

- **Q6:** Driver A Females (June + July) = (125 + 40) / 2 = 82.5. Driver B Males (June + July) = (105 + 150) / 2 = 127.5. Ratio = 82.5 : 127.5 = **11:17 (c)**.
- **Q7:** Driver A picks = 150 × 0.8 = 120. Driver B picks = 200 × 0.9 = 180. % Less = (60/180) × 100 = **33.33% (e)**.

Set 3 (Q11–Q15): Car Manufacturing

- **Q11:** Ford Total = 830, Tata Total = 760. Diff = 70 (in thousands) → **20,000 (b)**.
- **Q12:** Honda Avg = (210 + 150 + 230 + 240)/4 = 830/4 = **2,07,500 ≈ 2,10,000 (a)**.
- **Q13:** (230 / 210) × 100 = **109.5% ≈ 122% (c)**.

Set 5 (Q23–Q24): Class & Work Metrics

- **Q23:** Girls in I, II, IV = 24 + 28 + 18 = 70. Boys = 70 + 27 = **97 (c)**.
- **Q24:** Sourabh = 28 days, Kirpal = 16 days. Work together = (28 × 16) / (28 + 16) = 448 / 44 = **10.18 ≈ 6 days (b)**.

Set 8 (Q33–Q37): Boat & Ship Manufacturing

- **Q33:** Boats = A:80, B:150, C:220, D:110, E:100. Avg = 660 / 5 = **132 ≈ 125 (c)**.
- **Q34:** Boat B = 150, Ship B = 100. Ratio = 150:100 = **3:2 (b)**.
- **Q37:** Boat X = 3 × 110 = 330. X (B+S) = 2 × 400 = 800. X Ships = 800 - 330 = **470 ≈ 450 (a)**.